A LINEAR SERIES G^R_D YIELDS A G^S_D FOR EVERY S <= R

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the definition used is the in-source Definition definition-linear-series (navigation stub); the degree d is unchanged since the invertible module L is kept.

Editorial note. The label lemma-linear-series-trivial-existence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Linear series. Same unit, Definition definition-linear-series. The reduction uses the definition of a g^r_d as an equivalence class of pairs (L, V) from Definition definition-linear-series.

Lemma 0.1. *Let k be a field. Let X be a proper scheme of dimension ≤ 1 over k. If X has a $\mathfrak{g}^r_d$, then X has a $\mathfrak{g}^s_d$ for all $0 \leq s \leq r$.*

Proof. This is true because a vector space V of dimension $r + 1$ over k has a linear subspace of dimension $s + 1$ for all $0 \leq s \leq r$. □

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.